5(a)(i) Faraday's law states that the emf induced in a conductor is proportional to the rate of change of magnetic flux linkage.

(ii) The steel string near the permanent magnet gets magnetised (and produces its own magnetic field). [APPLICATION]

When the string vibrates, the magnetic flux density at the location of the coil changes. [APPLICATION]

There is a changing magnetic flux linkage with the coil [APPLICATION] so, according to Faraday's law [CONCEPT], an emf is induced in the coil [CONCLUSION].

(iii) Nylon string cannot be magnetised.

(b) The nodes of the stationary wave are located at the clamps, giving the first harmonic, so the length of 64 cm is equal to half a wavelength of the wave.

$\lambda/2 = 64$ cm
(need to explain properly and not just write the equation)

Frequency of the wave, $f = \dfrac{v}{\lambda} = \dfrac{300}{2(0.640)} = 234.4\ \text{Hz}$ [SHOW: GIVE MORE DP]

≈ 230 Hz

(c) maximum vibrational velocity of the string,

$$v_{\max} = \omega x_0 = (2\pi f)x_0 \text{ [CONCEPT]}$$
$$= (2\pi)(230)(1.50\times10^{-2})$$
$$= 21.68 \text{ m s}^{-1}$$

Maximum induced emf,
$$\varepsilon_{\max} = BLv_{\max} \text{ [CONCEPT]}$$
$$= (4.50\times10^{-3})(2.00\times10^{-2})(21.68)$$
$$= 1.95 \times 10^{-3} \text{ V or } 2.0 \times 10^{-3} \text{ V}$$

6(a)(i) As the half life of X (in years) is very long, it means that the decay constant is very